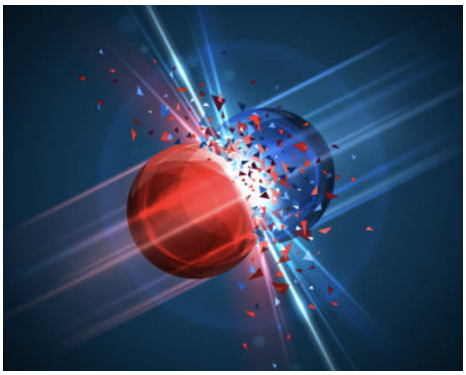


Figure 1:

Physics Lab Report

SPH4U7

Momentum Lab

Matthew Chen
matthew.chen26@ycdsbk12.ca

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1 Objective

The purpose of this lab is to observe the momentum of two objects during and after a collision. The lab combines several physics concepts, such as motion, elastic or inelastic collision, forces, dynamics, and uncertainty. The experiment consisted of two carts of different masses on either end of a track. They are gently pushed towards each other and repelled by the magnets at the point of collision. The individual mass of each cart is collected using a scale and Vernier Motion Detectors are used to collect the position-time data throughout the experiment.

2 Analysis

2.1 Is momentum conserved?

According to the Law of Conservation of Momentum, the total momentum of a system is conserved if there is no external force acting on it. To see if there is any external force in acting on the carts, we need to visualize the experiment.

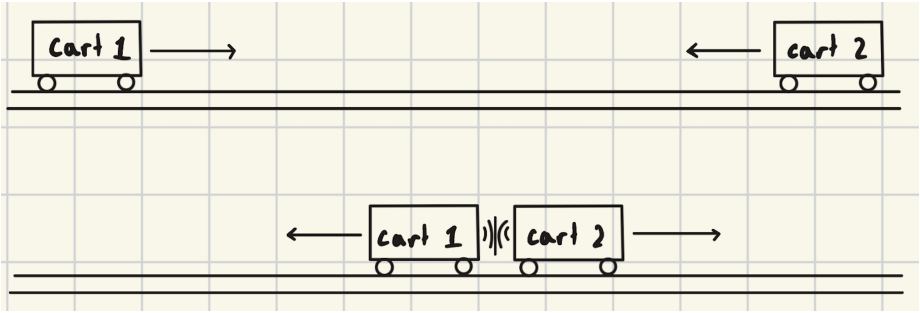


Figure 2: The two carts on the rail

In this experiment, friction is an external force acting between the wheels of the cart and the rail, so we expect a gradual loss of momentum over time. However, we need to consider whether the momentum is conserved across the brief duration of the collision. To qualitatively find out if there is a loss of momentum due to the collision, we need to examine the collision point (t_{impact}) by isolating the moments immediately before and after the collision. The data for the time before the crash is:

Time (s)	Total Momentum (kg·m/s)
4.30	0.198573056694
4.35	0.189700780500
4.40	0.169643836444
4.45	0.125669626250
4.50	0.0483518810833
4.55	-0.0450919519167
4.60	-0.119892086611
4.65	-0.160739918500
4.70	-0.177600455083
4.75	-0.184127449722
4.80	-0.185212301556
4.85	-0.184207359194
4.90	-0.184422953750
4.95	-0.184727337667
5.00	-0.184232369611

Table 1: Total Momentum of the System as a Function of Time

- **Step 1:** The magnitude of the initial momentum, before the crash at $t = 4.30\,\text{s}$ is:
$$P_{\text{initial}} \approx 0.199\ \text{kg} \cdot \text{m/s}.$$
- **Step B:** The magnitude of the final momentum, after the crash at $t = 5.00\,\text{s}$, is:
$$P_{\text{final}} \approx 0.184\ \text{kg} \cdot \text{m/s},$$
using the absolute magnitude for both initial and final to account for the sensor's coordinate flip.
- **Step C:** The percent difference is calculated as:
$$\%\ \text{Difference} = \frac{0.199 - 0.184}{0.1915} \times 100 \approx 7.83\%.$$

The total momentum before the collision was 0.199 and after was 0.184, resulting in a percent difference of roughly 7.83%. While the momentum dropped, this is consistent with our earlier statement about the presence of friction acting on the wheels of the cart. This difference is acceptable to conclude that momentum is effectively conserved during the collision event. Additionally, the use of repulsive magnets created an elastic collision, ensuring that the carts bounced cleanly without physical contact. This minimized the energy lost, as a result of heat and sound and allowed for a clearer analysis of the change in momentum.

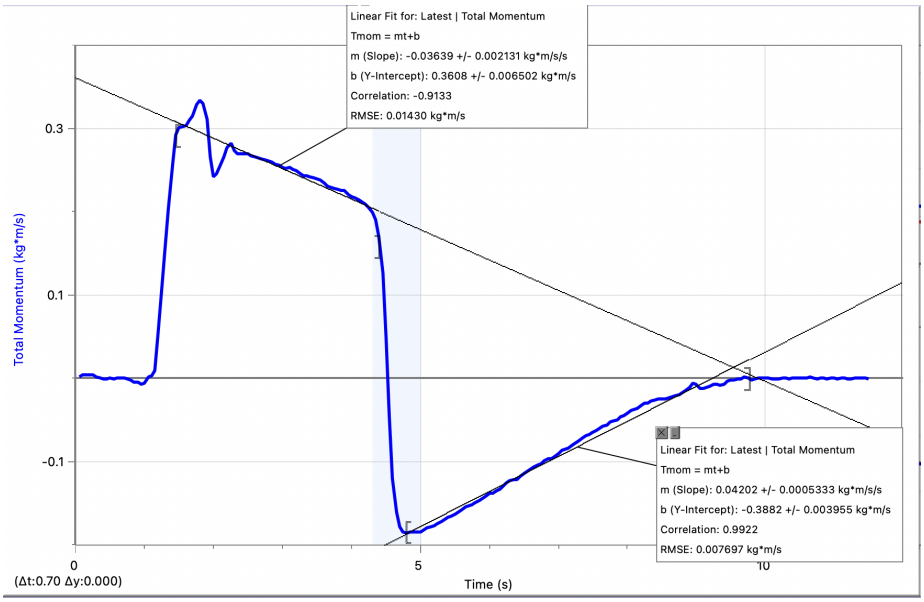


Figure 3: Momentum w.r.t. Graph

As seen in the momentum-time graph, the values are constantly decreasing in magnitude at an approximate constant rate. This decrease represents the change in momentum due to friction:

$$J = \Delta p.$$

The 7.83% loss we calculated is not a failure of the conservation law, but rather a measurement of the impulse of friction acting on the system during the interval of the collision. Therefore, the collision demonstrated the principle of conservation within the constraints of our environment, when friction is accounted for.

2.2 Finding Friction

Since our experiment tracks the motion of the carts before, during, and after the collision. The best way to calculate the average friction force generated throughout the experiment is by using the impulse equation. The impulse equation:

$$J = F_{net}\Delta t = \Delta p_{sys} \tag{1}$$

$J =$ impulse (N·s)
$F_{\text{net}} =$ net force acting on the object (N)
$\Delta t =$ time interval over which the force acts (s)
$\Delta p_{sys} =$ change in momentum of the system (kg·m/s)

The total momentum of the system is the momentum of both carts added together.

$$p_{sys} = p_1 + p_2 \tag{2}$$

The momentum of each cart can also be calculated, since mass times velocity at any point t is equal to the momentum at any point t. Our data from the experiment models velocity as:

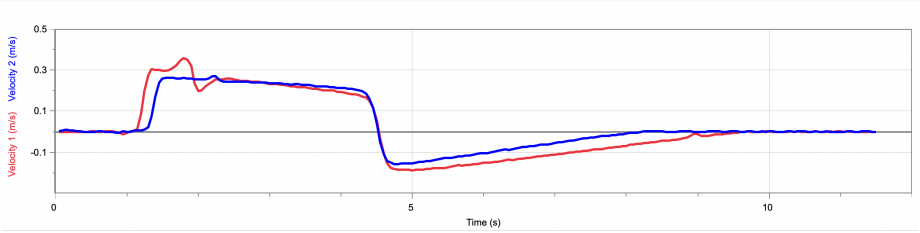


Figure 4: Velocity w.r.t.

With the help of Loggerpro, the velocity data and the mass data can calculate the momentum of the system over time. Since $p_{sys} = p_1 + p_2$, we can create a new calculated column in Loggerpro which shows the momentum over time. Given our masses:

Object	Mass (in g)	Uncertainty($\pm$ g)	Mass (in kg)
Cart 1	544.7	0.2	0.5447
Cart 2	534.2	0.2	0.5342

Table 2: Mass Table

The equation for the momentum in our experiment is:

$$0.5447v_1 + 0.5342v_2 = p_{at\ any\ point\ t} \tag{3}$$

Going back to Equation 1, the Law of Conservation of Momentum states that the total momentum of an isolated system with no external forces remains constant. However, in our experiment the momentum is not linear throughout since friction is acting between the wheels and the cart. The loss in momentum is attributed to the force acting against the carts, friction in this case. While there is a case to be made about the effect of drag on the system. Since drag $\propto v^2$ and the velocity is so little, the effect of drag on our experiment is negligible. So then the only force acting on the carts during the experiment is friction. With this new information, Equation 1 now looks like:

$$F_f \Delta t = \Delta p_{sys} \tag{4}$$

Which can be rearranged as:

$$F_f = \frac{\Delta p_{sys}}{\Delta t} \tag{5}$$

This equation can be arranged graphically with total system momentum as the y-axis and time as the x-axis. With this method, we can calculate the force of friction throughout the entire experiment, excluding the time of the collision. By excluding the time during the collision, there is two parts to this force of friction: before and after. Since the data for motion is directional, to calculate the average frictional force of the system we need to absolute value the total momentum. Therefore,

$$|\text{slope}| = F_{\text{friction}}.$$

2.3 Data

From the linear-fit outputs, we obtain:

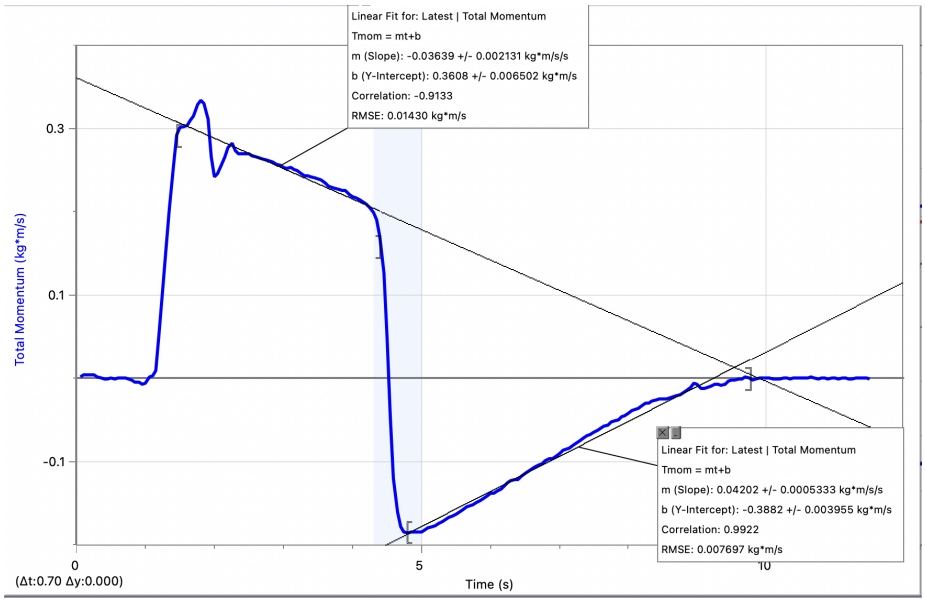


Figure 5: Momentum w.r.t. Graph with linear fit lines

- **Run 1 (Before Collision)** The carts have positive momentum, so friction acts in the negative direction:

$$F_{f1} = -0.03639 \text{ N}, \qquad \delta F_{f1} = \pm 0.002131 \text{ N}.$$

- **Run 2 (After Collision)** The carts now have negative momentum, so friction acts in the positive direction:

$$F_{f2} = 0.04202 \text{ N}, \qquad \delta F_{f2} = \pm 0.0005333 \text{ N}.$$

- **Calculating the Average Friction Force** Since friction depends only on the surface interaction and normal force, its magnitude should be identical in both directions. Therefore, we average the magnitudes before and

after the collision:

$$F_{\text{avg}} = \frac{|F_{f1}| + |F_{f2}|}{2}.$$

Substituting in our values to get:

$$\begin{aligned} F_{\text{avg}} &= \frac{|-0.03639| + |0.04202|}{2} \\ &= \frac{0.07841}{2} \\ &= 0.039205 \text{ N}. \end{aligned}$$

The uncertainty for the friction force is:

$$\delta F_{f1} = 0.002131 \text{ N}, \qquad \delta F_{f2} = 0.0005333 \text{ N}$$

1. Add the uncertainties:

$$\delta F_{\text{fsum}} = 0.002131 + 0.0005333 = 0.0026643$$

2. Divide by 2 to get the average uncertainty:

$$\delta F_{\text{avg}} = \frac{0.0026643}{2} = 0.00133215$$

Thus, the uncertainty in the average friction force is:

$$\boxed{\delta F_{\text{avg}} \approx 0.0013 \text{ N}}$$

Two sig. figs. are used for this uncertainty since the value is so small, only using 1 would be bit inaccurate.

Therefore, the final solution for the average friction force during the entire motion is $0.3920 \pm 0.0013N$

2.4 Uncertainty

Calculating uncertainty is critical in this experiment to validate the calculated frictional force. By considering the propagation of errors from both the scale and the random noise from the motion detectors, we can determine the reliability of our calculated friction value. This uncertainty analysis confirms that the observed loss in momentum is indeed within the expected range of frictional impulse, instead of being an anomaly caused by faulty equipment.

The total uncertainty in our experiment can be calculated with:

$$\%U_{total} = \%U_{mass} + \%U_{slope} \tag{6}$$

- **Calculate Mass % Uncertainty**

  Total mass of the system:

  $$M_{\text{sys}} = 544.7\,\text{g} + 534.2\,\text{g} = 1078.9\,\text{g}$$

  The total mass uncertainty of the system:

  $$\delta M_{\text{sys}} = 0.2 + 0.2 = 0.4\,\text{g}$$

  Finally, the percentage mass uncertainty:

  $$\%U_{\text{mass}} = \frac{0.4}{1078.9} \times 100 \approx 0.037\%.$$

- **Calculate Force % Uncertainty**

  Average force (taken from above):

  $$F_{\text{avg}} = 0.0392\,\text{N}$$

  Uncertainty from slope calculation (taken from above):

  $$\delta F_{\text{avg}} = 0.0013\,\text{N}$$

  Percentage for force uncertainty:

  $$\%U_{\text{slope}} = \frac{0.0013}{0.0392} \times 100 \approx 3.316\%.$$

- **Total Percentage Uncertainty**

  Add the percentage uncertainties:

  $$\%U_{\text{total}} = 0.037\% + 3.316\% = 3.353\%.$$

  Convert this to an absolute uncertainty in Newtons:

  $$\delta F_{\text{total}} = 0.0392 \times 0.03353 = 0.001314\ldots\ \text{N}$$

  After rounding the total uncertainty, the value remains $\pm 0.0013N$, the exact same uncertainty as the uncertainty as a result of random noise. This proves that the precision of the scale had almost no impact on the experiment and the uncertainty is almost entirely as a result of the noise in the motion sensor data.

3 Discussion

3.1 Frictional Force

The carts in this experiment experience low rolling friction for two key reasons. First, the wheels and track are both made from rigid materials, from the hard plastic wheels to the aluminum track that they roll on. Because neither surface significantly deforms under the weight of the cart, almost no energy is lost to hysteresis. This small deformation leads to an extremely small coefficient of rolling friction. Second, the rail constrains the motion of the cart to a perfectly straight path. Without this guiding structure, carts rolling freely on a flat table might drift sideways, causing the wheels to slide against perpendicular to the motion and generate additional friction.

3.2 Errors

Several sources of error influenced the accuracy of this experiment, and understanding their contributions is essential for evaluating the results. One significant limitation came from the ultrasonic motion detectors themselves. These sensors both have a minimum operating distance, less than 15 cm according to google. During the collision, the carts moved into the same region and it caused the ultrasonic pulses to overlap. This produced a sharp dip in the total momentum between t = 4.4s and t = 4.8s. Because this artifact does not represent real dynamics of the carts, only the before and after the collision segments of the experiment were used to determine average frictional force.

In addition to limitations as a result of the equipment, the experiment showed a clear systematic error caused by a slight tilt in the track. Analysis of the slopes before and after the collision showed a difference of approximately 0.0056N, which can correspond to a measurable gravitational component acting on the carts. This value is roughly four times larger than the random uncertainty associated with sensor noise ($\pm 0.0013N$), indicating that the uneven flooring and track was a dominant source of error. Therefore, improving the experiment would require prioritizing precise leveling of the track, through bubble levels or laser levels.